Absence of postprandial surge in coronary blood flow distal to significant stenosis: a possible mechanism of postprandial angina.

OBJECTIVES: This study was designed to investigate a possible mechanism of postprandial angina. BACKGROUND: Postprandial angina has been recognized for more than two centuries; however, its mechanism is still controversial. The most widely accepted mechanism involves increased myocardial oxygen demand after food intake. Recently, the redistribution in coronary blood flow (CBF) was suggested as a possible mechanism. METHODS: Twenty young, healthy volunteer controls and 20 patients with significant stenosis in the left anterior descending (LAD) or left main coronary artery were enrolled in the study. Coronary blood flow was evaluated in the distal LAD by using transthoracic Doppler echocardiography before and 15, 30, 45, and 60 min after food intake. In the CBF curve, the time velocity integral of diastolic flow (Dtvi) and the product of Dtvi and heart rate (HR) were measured. In six patients, these measurements were repeated after successful coronary intervention. RESULTS: In the healthy volunteer controls, Dtvi and Dtvi x HR increased after food intake with a peak value at 15 min, which indicates the presence of postprandial surge in the CBF. Fasting values and peak values at 15 min were significantly different (Dtvi: 15.1 +/- 4.9 cm/s vs. 18.9 +/- 5.9 cm/s, $p = 0.04$, Dtvi x HR: 862.2 +/- 261.5 cm/min vs. 1,174.2 +/- 307.5, $p = 0.002$). In contrast with the controls, despite postprandial increase in double product (HR x blood pressure), Dtvi and Dtvi x HR in the patient group decreased after food intake, with a nadir value at 45 min. Fasting values and nadir values at 45 min were significantly different (Dtvi: 24.0 +/- 19.6 cm/s vs. 19.3 +/- 17.1 cm/s, $p < 0.001$, Dtvi x HR: 1,449.6 +/- 1,044.0 cm/min vs. 1,273.4 +/- 1,000.9 cm/min, $p = 0.002$). In six patients, the CBF pattern resumed the normal pattern of postprandial surge in the CBF after successful coronary intervention. CONCLUSIONS: Results of our study suggest that "steal phenomenon" may play a role in the mechanism of postprandial angina.